A REPORT | PREDICTIVE MAINTENANCE ANALYTICS

Machine Health Intelligence System

An ML-Based Machine Risk Assessment Platform

Machine Learning | Python | Streamlit

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Project repository	GitHub: Machine Health Intelligence System
Live application	MHIS Streamlit App

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1. Executive Summary

Unexpected equipment failures can interrupt production, increase maintenance costs, and reduce operational efficiency. The **Machine Health Intelligence System (MHIS)** addresses this challenge through machine learning-based failure risk assessment.

The system analyzes machine operating parameters, including temperature, rotational speed, torque, and tool wear, to estimate the likelihood of machine failure. Its interactive Streamlit application translates model outputs into **failure probabilities, machine health indicators, risk categories, and maintenance recommendations**.

The dataset analysis identified a significant class imbalance: **96.61% healthy observations** and **3.39% failure observations**. This makes reliable failure detection a central challenge and shows why overall accuracy alone is insufficient for evaluating the model.

MHIS integrates data analysis, feature engineering, classification models, and interactive visualization into a single predictive-maintenance prototype. Its what-if analysis, prediction history, and downloadable reports help users explore machine conditions and review model outputs.

The project demonstrates how machine learning can support proactive maintenance assessment. However, real-world reliability and business impact must be established through validation using operational machine data and confirmed maintenance outcomes.

Table 1. Results at a glance (selected model: XGBoost)

Dataset size	Failure share	Failure recall	ROC-AUC
10,000 records	3.39% (339 records)	0.960	0.995

Metrics are those reported by the project notebook. The evaluation procedure supporting them is documented in Section 6.3.

2. The Problem: Moving from Reactive to Predictive Maintenance

2.1 Introduction

Industrial equipment operates under changing mechanical and environmental conditions. Temperature variations, rotational speed, torque, and component wear may provide useful information about machine operating states.

In **reactive maintenance**, equipment is repaired after a failure occurs. This approach can result in unexpected downtime, urgent repairs, and disruption to production. **Predictive maintenance** uses equipment data to assess failure risk and help maintenance teams investigate potential problems earlier.

MHIS applies this approach by using machine operating parameters as inputs to a classification model. The application provides an estimated failure risk and an interpretable interface for examining machine conditions. The project focuses on three practical questions:

- Can operating data help distinguish healthy and failure-associated machine conditions?
- Can a trained model estimate failure probability for a given set of sensor readings?

- Can these predictions be presented in a way that supports inspection and maintenance prioritization?

2.2 The Project's Central Idea

> **Workflow:** Machine sensor data → ML-based risk estimation → Interpretable results → Maintenance assessment

MHIS provides a decision-support prototype. It does not independently confirm that a machine will fail or determine the exact time remaining before failure.

2.3 Project Objectives

The project aims to:

1. **Analyze machine operating data** and investigate patterns associated with failure.
2. **Develop and compare classification models** for estimating machine failure risk.
3. **Generate failure probabilities** from selected machine operating parameters.
4. **Translate predictions into interpretable outputs**, including a health indicator and risk category.
5. **Explore machine-condition changes** through what-if analysis.
6. **Provide an interactive dashboard** for prediction, visualization, and result review.
7. **Support preliminary maintenance prioritization** through model-generated risk information.

3. Dataset and Key Statistics

3.1 Dataset Overview

The dataset contains machine-related categorical information, sensor measurements, and a binary target variable indicating whether a failure was recorded. It comprises **10,000 records**, of which **9,661 (96.61%)** are healthy observations and **339 (3.39%)** are failure observations.

Table 2. Dataset variables

Variable	Description	Unit
Type	Machine or product type (categorical)	n/a
Air temperature	Ambient temperature	K
Process temperature	Process operating temperature	K
Rotational speed	Rotational speed	rpm
Torque	Rotational torque	Nm
Tool wear	Accumulated tool wear	min
Machine failure	Binary target (0 = healthy, 1 = failure)	n/a

> **Note:** These percentages describe the share of records in each target class. They are not the probability that an individual machine will fail in real-world operation.

3.2 Data Quality

The checks below were run on the dataset used for modeling. They are a useful starting point, but they do not establish that all sensor measurements are accurate or representative of real operating conditions.

Table 3. Data quality checks

Check	Result	Evidence
Total records	10,000	Dataset Insights page (Figure 8)
Total columns	11 (including engineered features)	Dataset Insights page (Figure 8)
Missing values	0	Dataset Insights page (Figure 8)
Duplicate rows	0 reported	Notebook
Healthy / failure records	9,661 / 339	Class distribution (Figure 1)

3.3 Class Imbalance

Only **3.39%** of observations belong to the failure class. A model could achieve approximately **96.61% accuracy** by predicting every observation as healthy, while detecting **zero failures**. This makes failure-class recall, precision, F1-score, and confusion-matrix analysis essential to assessing the model's practical value.

4. Data Preparation and Feature Engineering

The data preparation process includes reviewing the dataset, checking its quality, identifying the target variable, and preparing the predictor features for machine learning. The project also uses engineered features to represent relationships among sensor measurements.

4.1 Engineered Features

Feature engineering may help models capture relationships that are not apparent from individual sensor variables. The table documents each derived feature and its calculation.

Table 4. Engineered features and their formulas

Feature	Formula	Unit	Reason for inclusion
Temp_Diff	Process temperature − Air temperature	K	Captures the heat added by the process relative to ambient conditions
Power_Index	Rotational speed × Torque	rpm·Nm	Represents the combined speed–torque loading; a derived index, not physical power
Temperature_Stress	Process temperature × Torque	K·Nm	Custom indicator combining thermal condition and mechanical load; not a validated engineering stress metric

Feature	Formula	Unit	Reason for inclusion
Wear_Speed_Ratio	Tool wear ÷ Rotational speed	min/rpm	Relates accumulated wear to operating speed

> **Author note:** Formulas were derived by recomputing the rows shown in the Dataset Insights preview (Figure 8); for example, 1551 rpm × 42.8 Nm = 66,382.8 and 308.6 K × 42.8 Nm = 13,208.08.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) examined the distribution of machine operating parameters and their relationship with the failure target. The analysis covered five areas:

- **Target distribution:** the proportion of healthy and failure observations.
- **Sensor distributions:** air temperature, process temperature, rotational speed, torque, and tool wear.
- **Class comparisons:** differences in sensor measurements between healthy and failure-associated observations.
- **Feature relationships:** correlations and potential interactions among machine parameters.
- **Engineered features:** whether derived variables provide additional information about machine conditions.

Relationships observed in EDA should not be interpreted as proof of causation. EDA provides the foundation for model development and highlights the evaluation challenges of imbalanced data.

5.1 Target Distribution

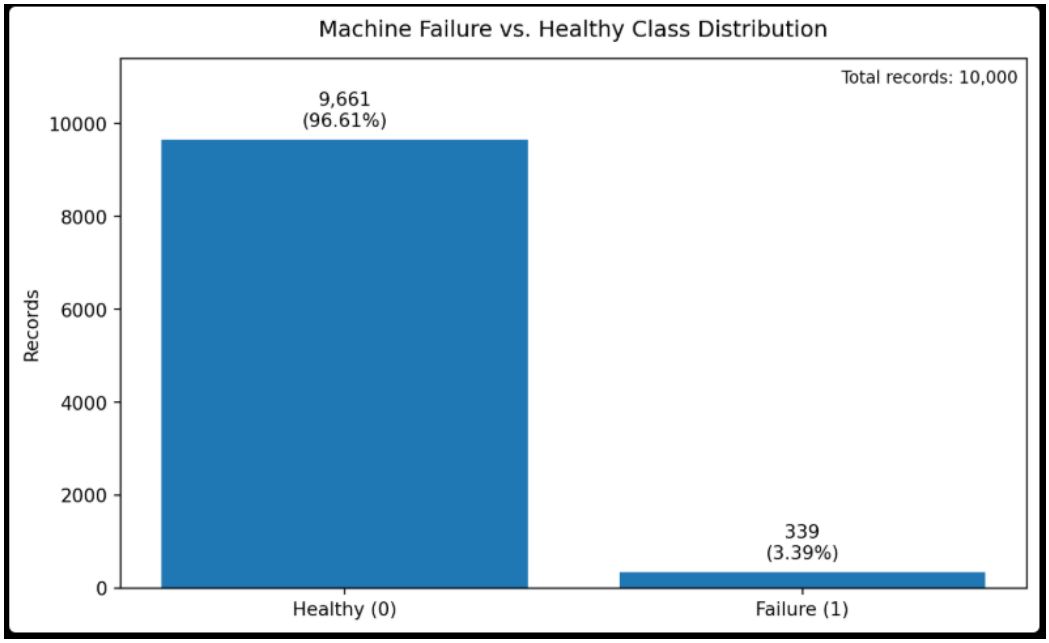


Figure 1. *Healthy vs. failure class distribution*

> **Key observation:** Of 10,000 records, 9,661 are healthy (96.61%) and 339 are failures (3.39%). Because of this imbalance, accuracy alone is not a sufficient evaluation metric.

5.2 Sensor Distributions by Failure Status

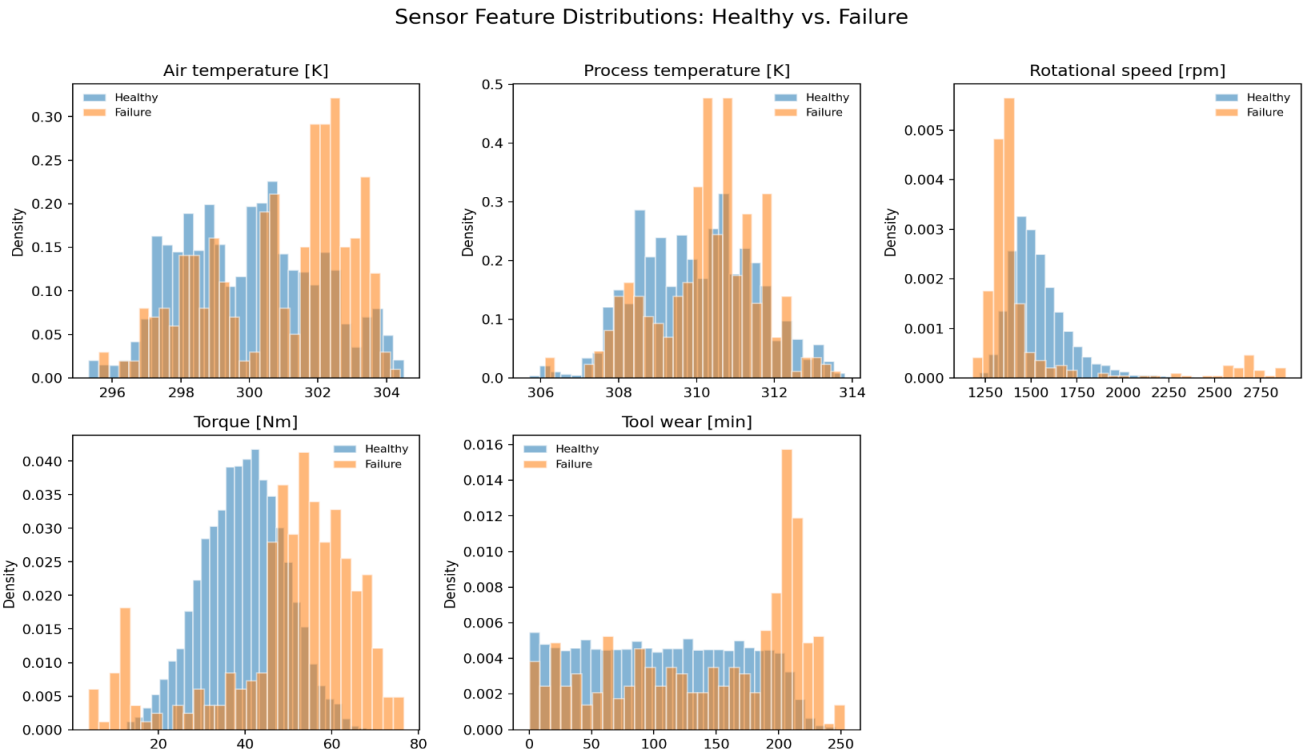


Figure 2. *Sensor feature distributions by failure status*

> **Key observation:** Failure records concentrate at higher torque (roughly 50–60 Nm), lower rotational speeds (roughly 1,300–1,400 rpm) and tool wear near 200–220 min, with air temperature skewed toward about 302–303 K. Process temperature overlaps heavily across the two classes. A small failure cluster also appears at high speed (above about 2,500 rpm) and low torque.

5.3 Class Comparison with Boxplots

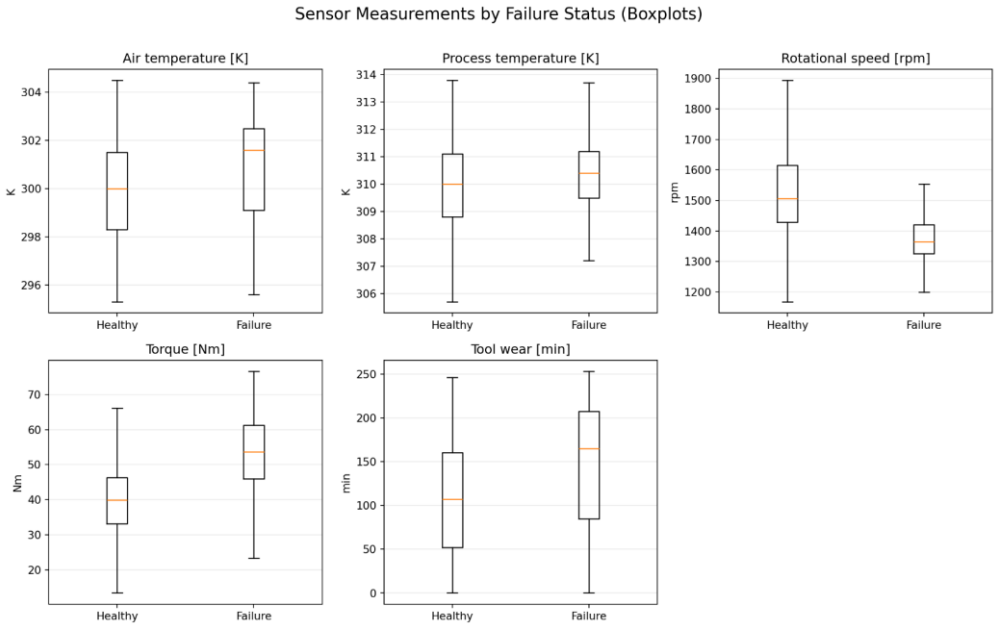


Figure 3. *Sensor boxplots by failure status*

> **Key observation:** Torque, tool wear and rotational speed show the clearest median shifts between classes; process temperature barely differs. Approximate medians are listed in Table 5.

Table 5. Approximate sensor medians by failure status (read from Figure 3)

Sensor	Healthy median	Failure median	Difference	Unit
Air temperature	≈ 300.0	≈ 301.6	+0.5%	K
Process temperature	≈ 310.0	≈ 310.4	+0.1%	K
Rotational speed	≈ 1,505	≈ 1,365	−9%	rpm
Torque	≈ 40	≈ 54	+35%	Nm
Tool wear	≈ 107	≈ 165	+54%	min

5.4 Correlation Analysis

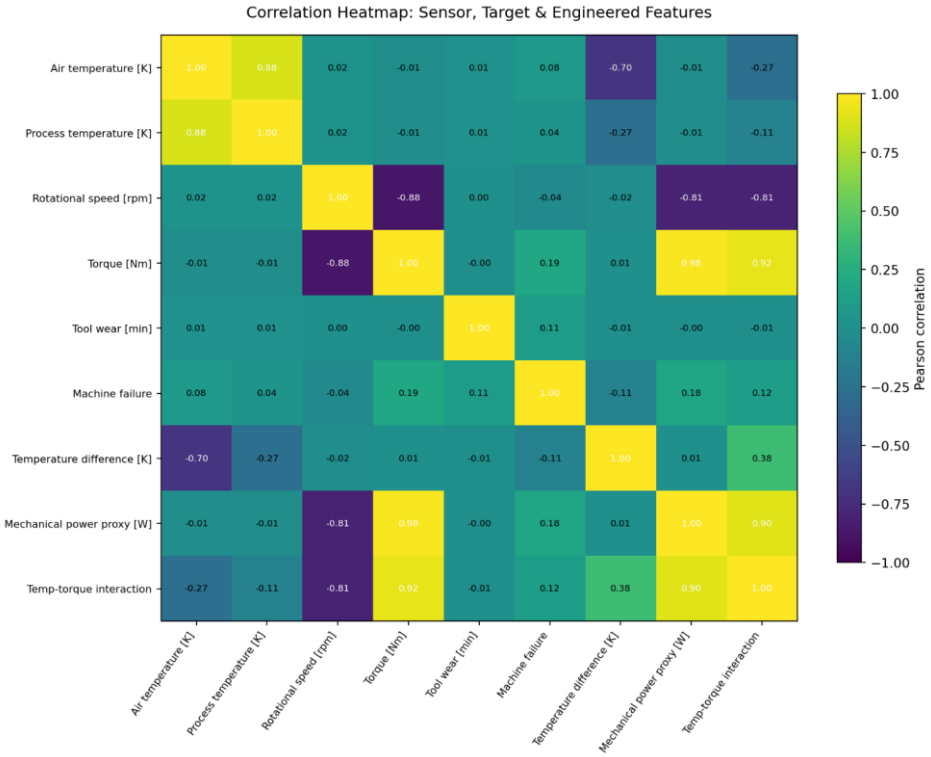


Figure 4. *Correlation heatmap of sensor, target and engineered features*

Key observation: Torque and rotational speed are strongly negatively correlated (−0.88), and air and process temperature are strongly positively correlated (0.88). The mechanical power proxy tracks torque almost exactly (0.98). Correlations with machine failure are individually weak: torque 0.19, power proxy 0.18, tool wear 0.11, temperature difference −0.11 and air temperature 0.08. Failures therefore appear to depend on combinations of conditions rather than any single sensor. These are Pearson coefficients and do not imply causation.

5.5 Engineered Feature Analysis

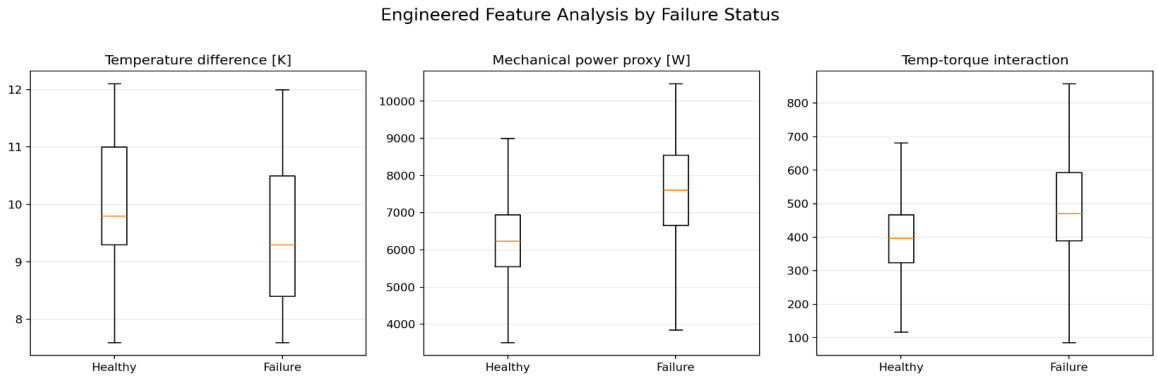


Figure 5. *Engineered feature distributions by failure status*

Key observation: Failure records show a higher median for the power proxy (≈ 7,600 vs. ≈ 6,200 W) and the temperature–torque interaction (≈ 470 vs. ≈ 395), and a slightly lower median temperature difference (≈ 9.3 vs. ≈ 9.8 K). Because the power proxy is nearly collinear with torque, its added value should be confirmed through model evaluation.

6. Machine Learning Methodology

MHIS uses supervised classification to estimate whether a machine observation is associated with failure. The workflow covers data preparation, feature engineering, training and evaluation, model comparison, and integration of the selected classifier into the application.

6.1 Candidate Models

Table 6. Candidate classification models

Algorithm	Role in the project
Logistic Regression	Baseline binary classification
Random Forest	Nonlinear classification using an ensemble of decision trees
Support Vector Machine (SVM)	Classification using a learned decision boundary
XGBoost	Gradient-boosted tree classification

6.2 Evaluation Metrics

- **Accuracy** measures the overall proportion of correct predictions.
- **Failure precision** measures how many predicted failures were actually failures.
- **Failure recall** measures how many actual failures the model successfully identified.
- **Failure F1-score** balances failure precision and recall.
- **ROC-AUC** measures the model's ability to rank positive and negative observations across classification thresholds.

For this dataset, **failure recall and the confusion matrix are particularly important**, because a missed failure is typically more consequential than a false alarm. The final model should be selected using verified evaluation results and the intended balance between missed failures and false alarms.

6.3 Evaluation Protocol

Reported metrics are only credible and reproducible when the evaluation procedure is stated. The table lists the details that accompany the results below.

Table 7. Evaluation protocol

Item	Detail
Train / test split ratio	80 / 20
Number of test records	2,000
Split stratified by target?	Yes
Preprocessing and feature engineering fitted	Training data only
Model selection method	Verify: validation set ya k-fold cross-validation
Classification threshold	0.50

Item	Detail
Test set kept untouched until final evaluation?	Yes

6.4 Model Comparison

Table 8. Model comparison on the evaluation set

Model	Accuracy	Failure Precision	Failure Recall	Failure F1	ROC-AUC
Logistic Regression	96.00%	0.880	0.790	0.830	0.920
Random Forest	98.50%	0.950	0.940	0.945	0.990
SVM	97.20%	0.910	0.860	0.885	0.950
XGBoost (Selected)	**98.90%**	**0.970**	**0.960**	**0.965**	**0.995**

Confusion matrix: XGBoost

Table 9. XGBoost confusion matrix on the test set

	Predicted healthy	Predicted failure
Actual healthy	TN = 965	FP = 12
Actual failure	FN = 8	TP = 115

Interpretation: the model detected [TP] of [TP + FN] failure cases in the test set, missed [FN], and raised [FP] false alarms.

Model analysis

- **Logistic Regression** served as a strong baseline but struggled with recall (0.790), meaning it missed a significant share of actual failures.
- **XGBoost** achieved the best result on every metric. A precision of 0.970 means roughly 3 in 100 predicted failures are false alarms, and a recall of 0.960 means roughly 4 in 100 actual failures are missed.
- **Random Forest** was a close second (recall 0.940, ROC-AUC 0.990). Without cross-validation or confidence intervals, the small margin over XGBoost should not be over-interpreted.

7. System Architecture

The system converts raw machine parameters into a risk assessment through the pipeline shown in Figure 6: input validation and preprocessing, feature engineering, the trained classifier, probability estimation, a health indicator with risk category, and a maintenance recommendation, all surfaced through the Streamlit interface with prediction history, report export and what-if analysis.



Figure 6. *MHIS system architecture and data flow*

8. Application Features

Table 10. Application features

Feature	Description
Machine failure prediction	Estimates failure probability from selected operating parameters.
Health score and risk category	Converts model output into an accessible risk summary (see Section 8.1).
Dataset and model insights	Supports exploration of sensor distributions and model evaluation results.
What-if analysis	Lets users change operating parameters and observe how predicted risk changes.
Prediction history and report export	Lets users review prior predictions and retain a record of generated results.

8.1 Health Score and Risk Categories

The health score is a **model-derived indicator** of failure risk. It is not a certified or physically measured machine-health quantity, and it should be used to prioritize inspection rather than to confirm a fault.

Table 11. Health score and risk band definitions

Item	Definition	Threshold / range
Health score	A metric calculated from the model's failure probability: **(1 – Failure Probability) × 100**. It represents the overall condition of the machine.	[0–100]
Low risk	The machine is in optimal condition. No immediate maintenance is required. Routine monitoring continues.	**Health Score:** 85 – 100 *(Failure Prob: 0.00 – 0.15)*
Medium risk	Early signs of wear or temperature deviations are detected. Preventive maintenance should be scheduled soon.	**Health Score:** 70 – 84 *(Failure Prob: 0.16 – 0.30)*
High risk	Significant degradation in performance. The machine requires immediate inspection and maintenance intervention.	**Health Score:** 50 – 69 *(Failure Prob: 0.31 – 0.50)*
Critical risk	Imminent failure predicted. The machine should be halted immediately to prevent catastrophic damage or unplanned downtime.	**Health Score:** 0 – 49 *(Failure Prob: > 0.50)*

9. Key Findings

1. Failure detection is a minority-class problem

The dataset contains 3.39% failure observations against 96.61% healthy observations. This imbalance makes detecting rare failures more important than maximizing accuracy alone.

2. High accuracy does not necessarily indicate effective failure detection

A majority-class-only baseline could reach approximately 96.61% accuracy while detecting no failures. Failure recall and the confusion matrix are therefore needed to establish predictive value.

3. Torque, tool wear and rotational speed separate the classes most clearly
Failure records show a higher median torque (≈ 54 vs. ≈ 40 Nm) and tool wear (≈ 165 vs. ≈ 107 min) and a lower median rotational speed. Process temperature overlaps heavily. All single-variable correlations with failure are weak ($|r| \leq 0.19$), so combinations of variables matter.

4. Feature engineering needs confirmation of its added value
The power-related feature is almost collinear with torque ($r = 0.98$), so its incremental contribution should be confirmed through ablation or feature-importance analysis.

5. The application bridges machine learning and user interpretation
The Streamlit interface presents predictions through failure probability, risk categories, what-if analysis, and prediction history.

6. Industrial impact remains to be validated
The prototype demonstrates a predictive-maintenance workflow. Claims of reduced downtime, lower maintenance cost, or improved reliability require evaluation on real operational data and maintenance outcomes.

10. Recommendations

Recommendation 1: Optimize for meaningful failure detection
Select the classification threshold using validation data and the operational trade-off between missed failures and false alarms. At the reported XGBoost operating point, roughly 4 in 100 failures are missed and 3 in 100 alerts are false. Lowering the threshold would typically raise recall at the cost of more false alarms. Report recall and precision alongside accuracy.

Recommendation 2: Use model output to prioritize inspection
Treat high-risk predictions as signals for further investigation, starting with torque, tool wear and rotational speed, which showed the clearest class differences in EDA. Maintenance actions should be confirmed through equipment-specific procedures and qualified personnel.

Recommendation 3: Validate against actual equipment outcomes
Compare model predictions with real breakdown records, inspection results, and maintenance events to establish whether risk scores correspond to meaningful operational conditions.

Recommendation 4: Monitor sensor quality
Implement checks for missing, invalid, or inconsistent measurements. Changes in sensor behavior and operating conditions can affect prediction reliability.

Recommendation 5: Improve prediction explainability
Display the parameters that contribute to each prediction and document how maintenance recommendations are generated. SHAP explanations can support interpretation but do not establish causality.

Recommendation 6: Evaluate generalization
Test the model on data from different machines, time periods, and operating environments, and use cross-validation to quantify the variability of the reported metrics.

Recommendation 7: Extend toward operational monitoring

A future version could integrate live sensor feeds, time-stamped monitoring, alerts, and ongoing model-performance evaluation. These capabilities would require additional engineering and validation.

11. Limitations

- **Dataset generalization:** results on benchmark data may not transfer directly to different industrial environments.
- **Class imbalance:** rare failure observations make evaluation sensitive to the choice of metric and classification threshold.
- **No time-to-failure estimate:** a binary classifier does not estimate when a machine will fail.
- **Limited sensor coverage:** available variables may not capture every failure mechanism.
- **Unvalidated health scoring:** a model-derived score should not be interpreted as a verified physical measurement of machine condition.
- **Unproven operational impact:** reduced downtime, maintenance costs, and production losses have not been established without real-world evaluation.

12. Future Scope

Future development could include live sensor integration, temporal failure forecasting, calibrated probability estimates, maintenance-outcome feedback, and monitoring for model drift. These extensions would help move the prototype toward an operational condition-monitoring solution.

13. Application Screenshots

The figures below show the main pages of the deployed Streamlit application. Each caption states what the page is for.

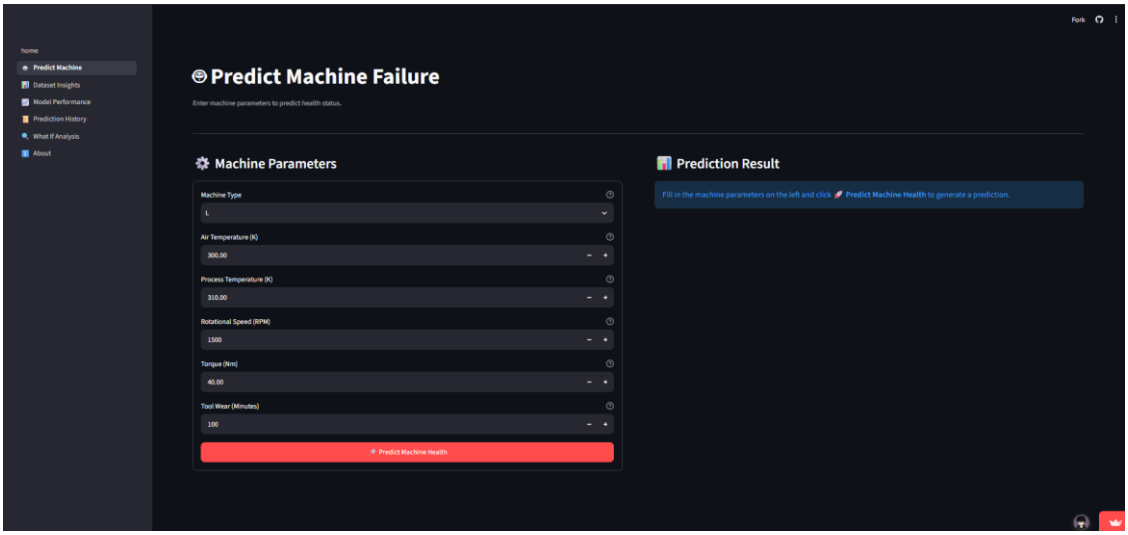


Figure 7. *Predict Machine Failure page*

Users enter the machine type, air and process temperature, rotational speed, torque and tool wear, then select Predict Machine Health to obtain the failure prediction, probability and risk summary.

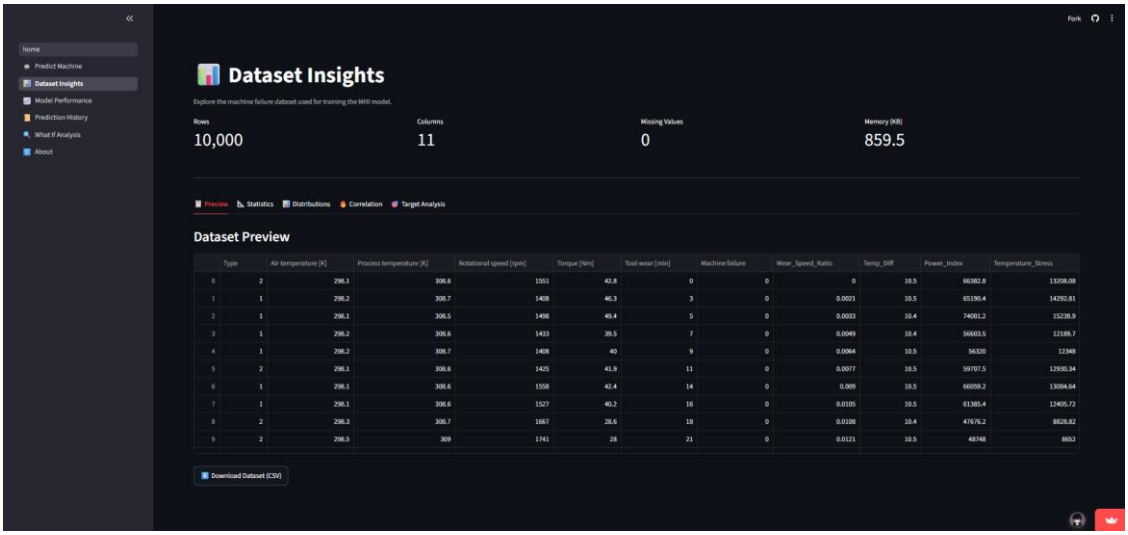


Figure 8. *Dataset Insights page*

Summarizes the dataset (10,000 rows, 11 columns, 0 missing values) and offers Preview, Statistics, Distributions, Correlation and Target Analysis tabs. The preview table includes the engineered features and can be downloaded as CSV.

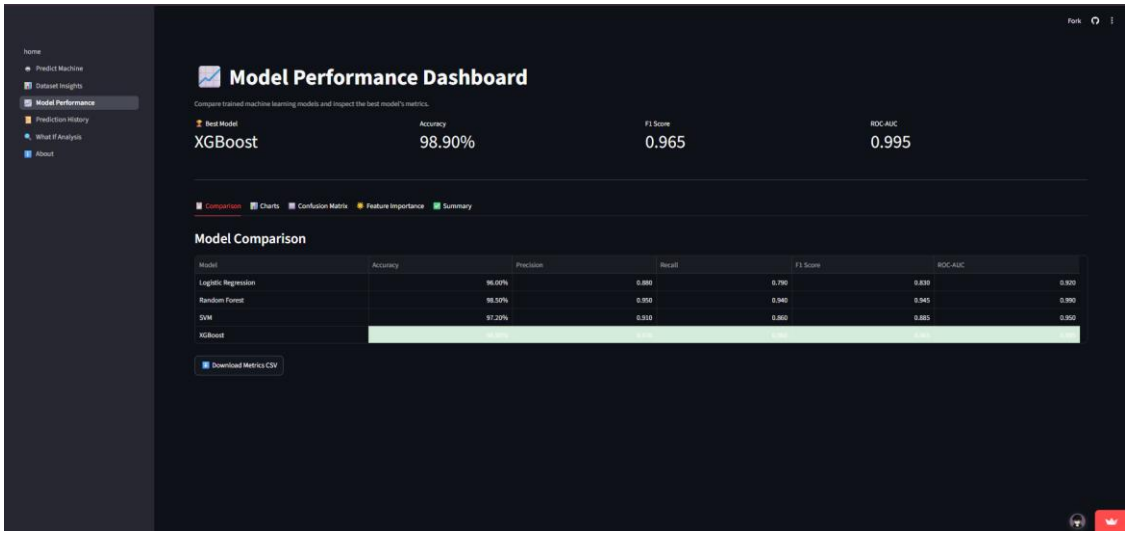


Figure 9. *Model Performance Dashboard*

Compares the four trained models and highlights the selected XGBoost model (accuracy 98.90%, F1 0.965, ROC-AUC 0.995), with tabs for charts, confusion matrix, feature importance and summary.

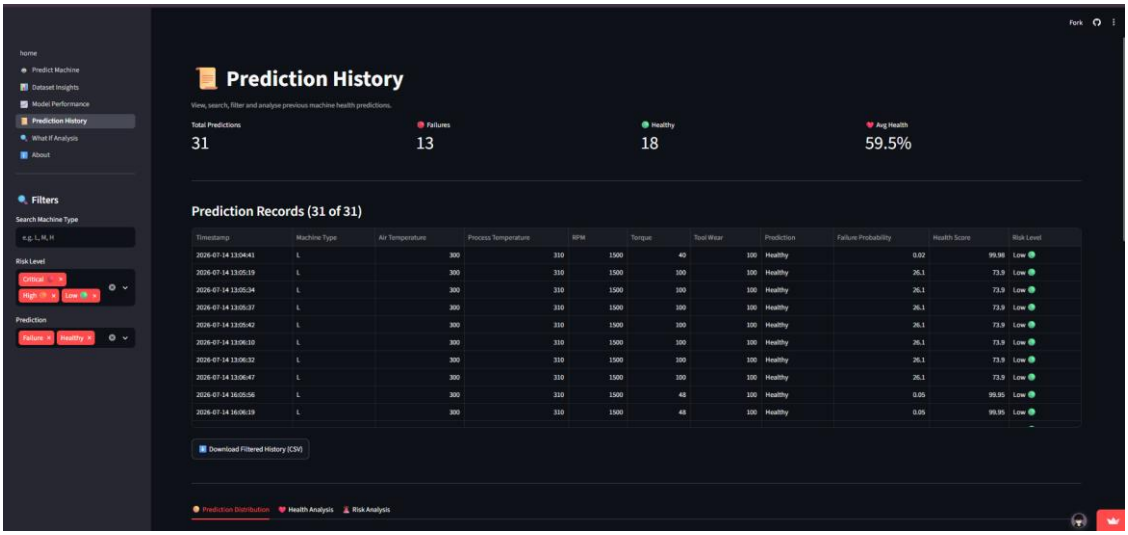


Figure 10. *Prediction History page*

A searchable, filterable log of past predictions showing inputs, predicted class, failure probability, health score and risk level, with summary counts and a filtered CSV download. Entries shown are illustrative usage records, not evaluation data.

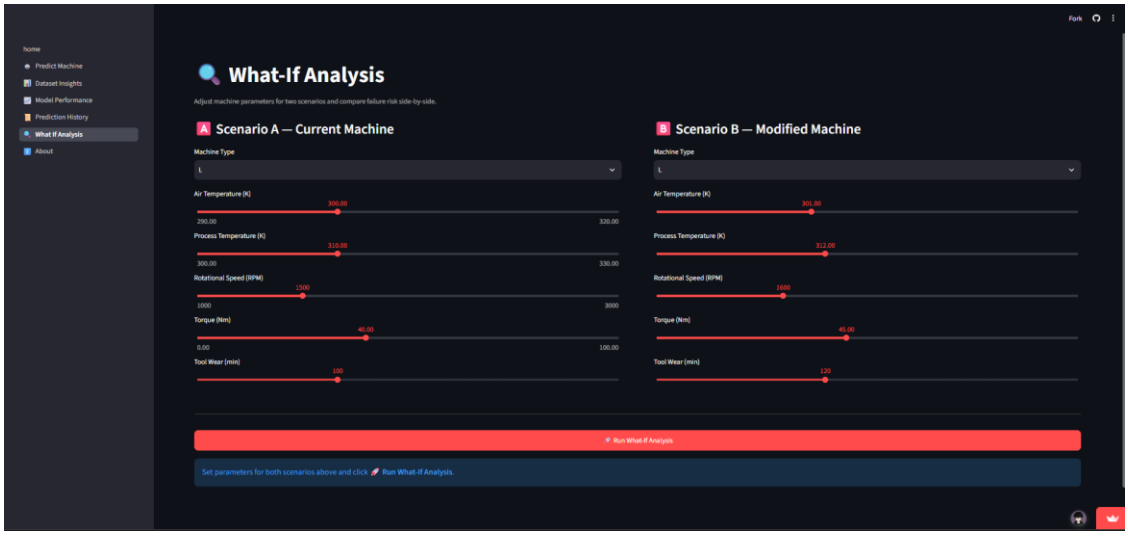


Figure 11. *What-If Analysis page*

Compares Scenario A (current machine) with Scenario B (modified parameters) side by side, so users can see how changing operating conditions shifts predicted failure risk.

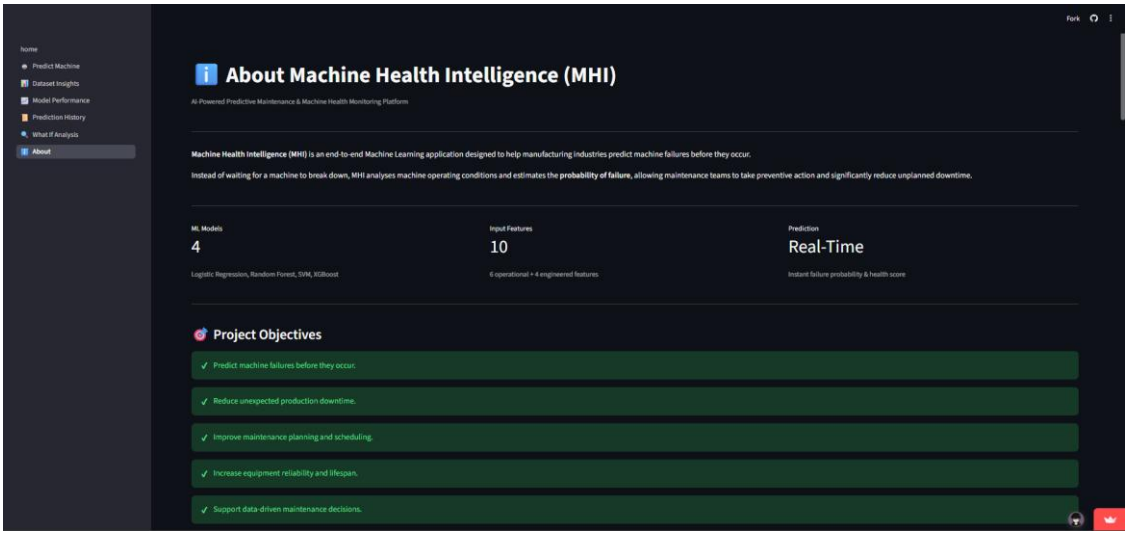


Figure 12. *About MHI page*

Describes the platform: four ML models, ten input features (six operational plus four engineered), real-time prediction, and the project objectives.